



Product Configuration Guide

PRODUCT CONFIGURATION GUIDE



MODEL NO/DESCRIPTION

Product Name: Bluetooth® Smart IoT Gateway
(Connection Ver.)

Model: G1-E-grapes

Version	V001
release time	2023-06-12

Table of contents

Preface	1
1. Basic Configuration	1
1.1 Configuration Method	1
1.2 Login Password of Configuration Page	5
1.3 AP Configuration	5
2. Network Configuration	9
2.1 Ethernet Connection	10
2.2 Wireless connection	13
3. Data Collection & Service Access	20
3.1 Data Collection	20
3.2 Service Access	21
3.3 Data Filtering	30
3.4 Scan Parameter Configuration	31
3.5 Remote Control (Only MQTT)	33
3.6 Disconnected Storage	33
4. Other Configuration	33
4.1 Top Lights Settings	33
4.2 Time Configuration	35
4.3 Automatic Management	37
4.4 Firmware Upgrade	38
4.5 Troubleshooting	46
4.6 Restart the Gateway	48
4.7 Factory Reset	49
4.8 Brick Rescue Mode	49
5. Statement	51

Preface

This specification applies to the G1-E-grapes model gateway.

revision	date	author	describe
V001	2023.06.12	Yancy	First draft

1. Basic Configuration

The gateway device can be configured through a web configuration interface.

1.1 Configuration Method

There are two ways to enter the web configuration page of the gateway.

1.1.1 Use AP mode to enter the configuration page

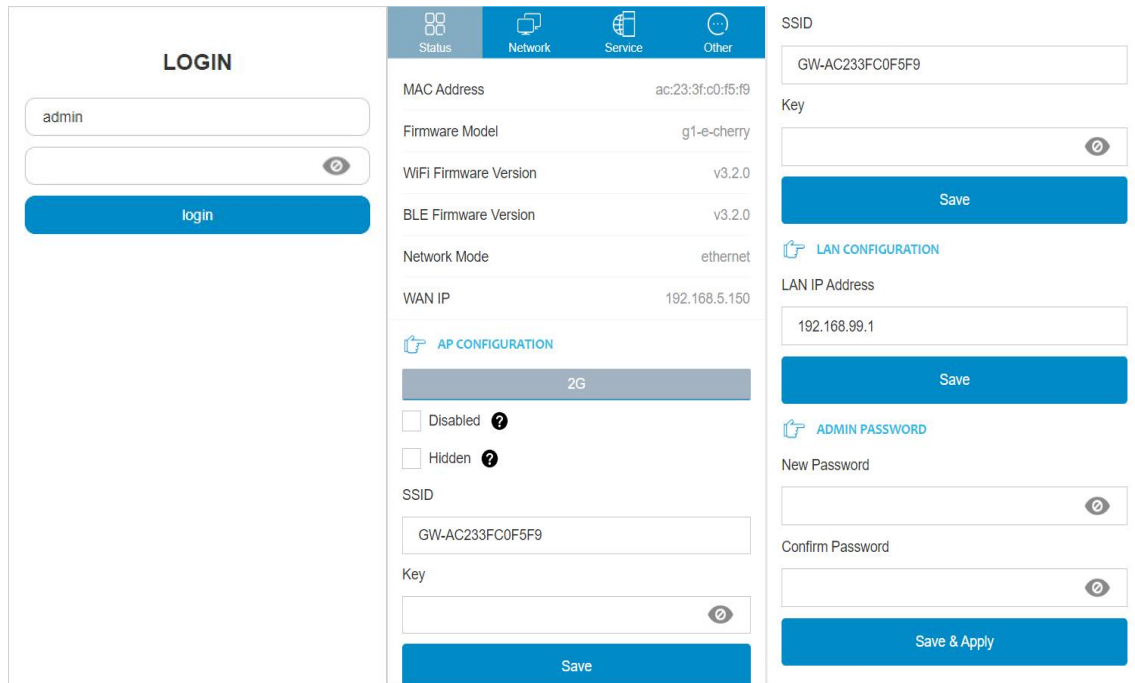
AP mode is only used for gateway configuration, and will not affect the current networking mode of the gateway as a client.

AP mode provides WiFi hotspot with SSID name "GW-XXXXXXXXXXXX", where XXXXXXXXXXXX is the hexadecimal uppercase characters corresponding to the gateway MAC address. Users can scan and connect with a computer or mobile phone. By default, no password is required to connect to this WiFi hotspot. In order to control access rights, you can modify the AP SSID name in the WEB configuration interface, and the IP address needs to be set to the format of 192.168.X.1 (the subnet segment of the gateway cannot be the same as the network segment of the access route ¹), and set the WiFi connection password.

After connecting to the gateway WiFi, use a computer or mobile phone browser to open the link:

¹ Network segment: For example, IP: 192.168.X.1, when the X is the same, the network segment is the same, and when the X is different, the network segment is different.

<http://192.168.99.1> to enter the login interface of the configuration interface (see the figure below).



The screenshot displays the MINNEW configuration interface. On the left is a 'LOGIN' section with a username field (containing 'admin'), a password field, and a 'login' button. The main area is divided into four tabs: 'Status', 'Network', 'Service', and 'Other'. The 'Network' tab is active, showing fields for 'MAC Address' (ac:23:3f:c0:f5:f9), 'Firmware Model' (g1-e-cherry), 'WiFi Firmware Version' (v3.2.0), 'BLE Firmware Version' (v3.2.0), 'Network Mode' (ethernet), and 'WAN IP' (192.168.5.150). Below these are expandable sections for 'AP CONFIGURATION' (showing '2G' mode and checkboxes for 'Disabled' and 'Hidden'), 'LAN CONFIGURATION' (showing 'SSID' as 'GW-AC233FC0F5F9' and 'Key' as an empty field), and 'ADMIN PASSWORD' (showing 'New Password' and 'Confirm Password' fields). Each configuration section has a 'Save' button, and the 'ADMIN PASSWORD' section has a 'Save & Apply' button.

The interface fields are described in detail below.

field	name	describe
MAC Address	Gateway MAC	The gateway MAC address in hexadecimal lowercase characters.
Firmware Model	Firmware model	The firmware model of the gateway
WiFi Firmware Version	WiFi firmware version	The firmware version number of the WiFi in the gateway
BLE Firmware Version	BLE firmware version	The firmware version number of BLE in the gateway
Network Mode	network mode	Displays the mode in which the gateway connects to the network. There are two modes ethernet. When the wireless is not connected, it is displayed as N/A.

WAN IP	Internet IP address	The IP obtained by connecting to the external network as a client is displayed as N/A when not connected
Disabled	Whether to turn off WiFi	Turn off or turn on WiFi hotspot
Hidden	Whether to hide WiFi	Hide or show WiFi hotspots
SSID	AP SSID	Configure the SSID in AP mode
Key	AP password	Configure the connection password in AP mode, more than 8 passwords
LAN IP Address	AP intranet IP address	Configure the IP in AP mode, the default is 192.168.99.1
New Password	Set a new login password	Login password for configuring the gateway configuration page
Confirm Password	confirm the new login password	Login password for configuring the gateway configuration page

1.1.2 Enter the Configuration Page In The LAN

As long as the devices with browsers such as computers and mobile phones are in the same LAN as the gateway, the computer can enter the configuration page of the gateway by entering the IP address in the browser instead of connecting to the WiFi hotspot of the gateway. It's just that the IP address is no longer 192.168.99.1, and the IP address obtained by the gateway needs to be scanned by the ARP scanning tool.

View through router

Some enterprise-level routers can view the ARP list (the list of IP addresses and MAC addresses) on the management page of the router.

View from a Linux computer

- Take Ubuntu as an example, install the *arp-scan* command

\$ sudo apt-get install arp-scan

- Run the command to view the IP-MAC list

\$ sudo arp-scan -l

- Run the command to filter out the MAC address of the gateway

\$ sudo arp-scan -l | grep "ac:23"

```
lyx@ubuntu:~/openwrt/openwrt_thingoo_g1-c$ sudo arp-scan -l | grep "ac:23"
192.168.0.90      ac:23:3f:c0:00:be      (Unknown)
192.168.0.122    ac:23:3f:c0:07:d6      (Unknown)
192.168.0.140    ac:23:3f:c0:00:f4      (Unknown)
```

View from a Windows computer

Download the arp-scan tool from github <https://github.com/QbsuranAlang/arp-scan-windows>

If your LAN is 192.168.0.1, the subnet mask is 255.255.255.0

Run the command as follows:

arp-scan -t 192.168.0.1/24

The meaning here is to scan the LAN IP: 192.168.0.x (from 192.168.0.1 to 192.168.0.255). 24 means the number of binary 1s in the subnet mask (255.255.255.0).

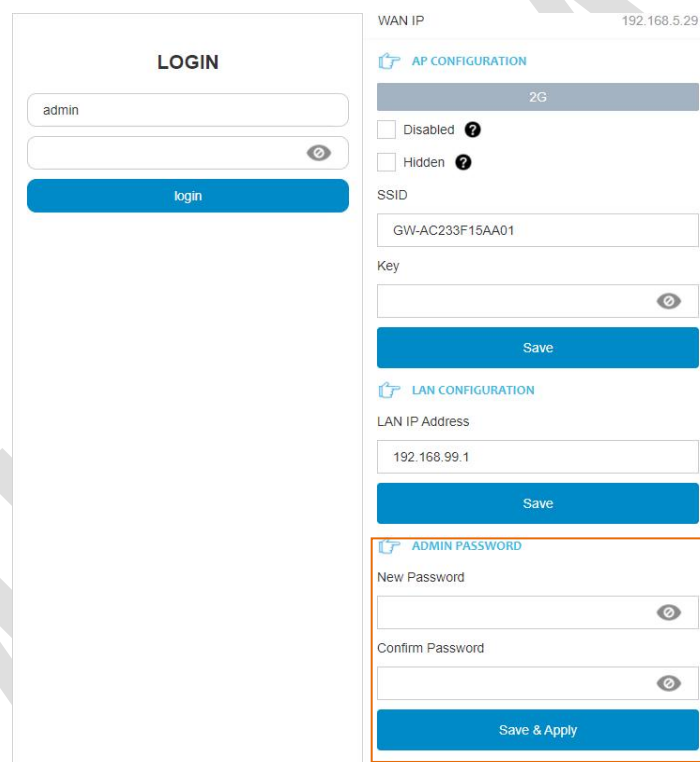
```
C:\Users\Administrator\Desktop\arp-scan-windows-master\arp-scan-windows-master\arp-scan\Release(x86)>arp-scan.exe
Usage: arp-scan.exe -t [IP/slash] or [IP]

C:\Users\Administrator\Desktop\arp-scan-windows-master\arp-scan-windows-master\arp-scan\Release(x86)>arp-scan.exe -t 192.168.0.1/24
Reply that DC:FE:18:25:F8:A8 is 192.168.0.1 in 0.881312
Reply that 4C:CC:6A:8E:8E:3D is 192.168.0.3 in 11.043234
Reply that 0C:4B:54:D8:83:F5 is 192.168.0.4 in 11.794980
Reply that 94:D8:B8:2B:E5:C3 is 192.168.0.5 in 11.415704
Reply that 94:D8:B8:2F:FE:0A is 192.168.0.6 in 10.672595
Reply that 34:96:72:FB:05:4F is 192.168.0.7 in 0.783680
Reply that E0:62:90:32:E0:02 is 192.168.0.8 in 10.805041
Reply that 1C:87:2C:77:8E:98 is 192.168.0.13 in 0.898064
Reply that 2C:4D:54:57:10:30 is 192.168.0.20 in 0.047900
Reply that 24:69:68:E4:B9:81 is 192.168.0.21 in 12.689903
Reply that 50:FA:84:83:45:E4 is 192.168.0.26 in 0.885762
Reply that 00:CF:E0:32:4B:97 is 192.168.0.35 in 1.220540
```

View through the batch configuration assistant developed by our company

The batch configuration assistant is a tool for batch configuration based on the local area network. It can scan the IP address of the gateway in the local area network. If necessary, you can contact our sales team.

1.2 Login Password of Configuration Page



The screenshot displays the gateway configuration interface. On the left is a 'LOGIN' panel with a text input for 'admin', a password input field, and a 'login' button. On the right is the main configuration area. At the top, it shows 'WAN IP' as '192.168.5.29'. Below this are three sections: 'AP CONFIGURATION' with a '2G' toggle, 'Disabled' and 'Hidden' checkboxes, and an 'SSID' field containing 'GW-AC233F15AA01'; 'LAN CONFIGURATION' with a 'LAN IP Address' field containing '192.168.99.1'; and 'ADMIN PASSWORD' which is highlighted with an orange border. The 'ADMIN PASSWORD' section includes 'New Password' and 'Confirm Password' fields, both with eye icons, and a 'Save & Apply' button.

The default login password on the gateway configuration page is empty, and you can log in directly by clicking login. For security, please set the administrator login password in ADMIN PASSWORD on the "**status**" page.

1.3 AP Configuration

The hotspot AP of the gateway WiFi is always on by default and does not require a password to connect. You can hide or close the AP hotspot, and you can also set the connection password in AP mode. There are two ways to control the AP of the gateway WiFi: through the reset button of

the gateway or through the web page configuration.

Note: The shutdown of the WiFi hotspot will not affect the current gateway's networking method as a client.

1.3.1 Turn Off or Turn On The AP

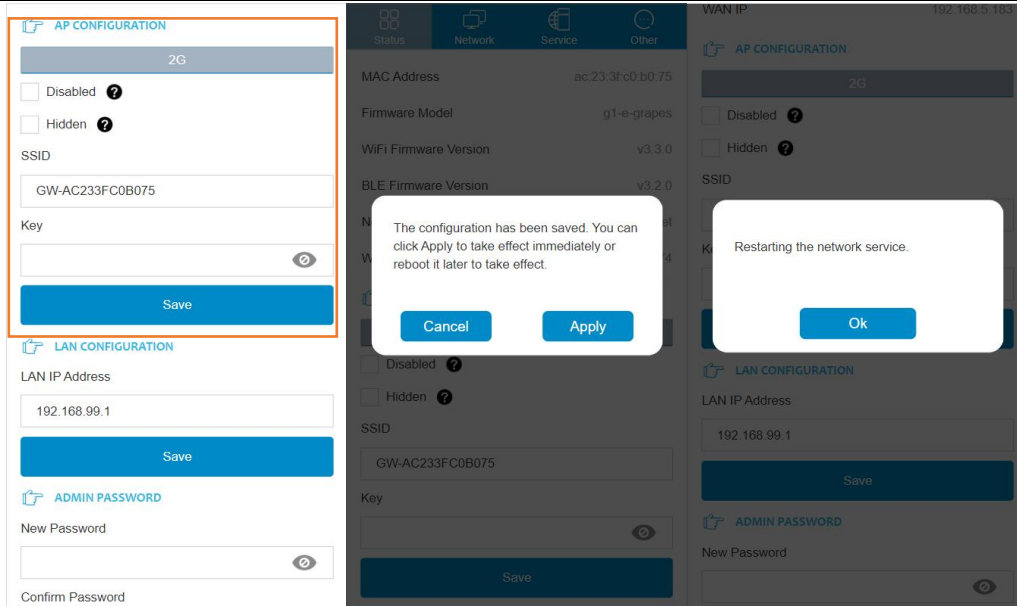
1. Reset key:

In the fully activated state, that is, when the gateway lights are dynamically bright, use a sharp object to press and hold the reset button of the device through the reset hole for **less than 2 seconds** and then release it, which will turn on or off the hotspot AP of the gateway WiFi.

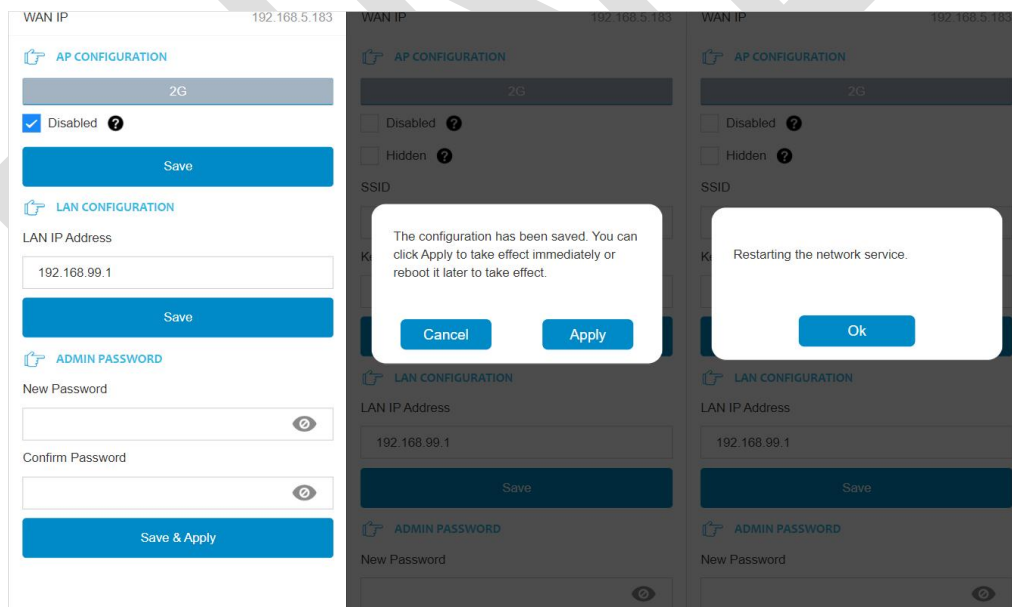
Note: Press and hold the Reset button for no more than two seconds, and the factory settings will be restored when the time exceeds 5 seconds.

2. Web page configuration:

Close the AP : Check the **"Disabled"** option in the **"Status"** page , click the **"Save"** button, click the **"Apply"** button, a pop-up will ask whether to restart the gateway service immediately (see the figure below), click **"Apply"** to take effect immediately , and then click **"OK"**. Then refresh the page, wait for the gateway service to restart and update the interface information, as shown in the following figure.

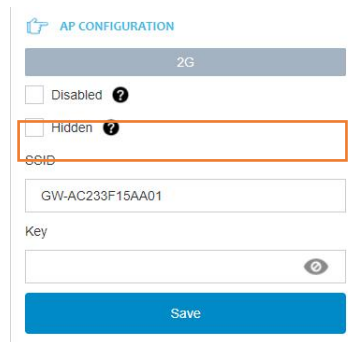


Enable AP : Uncheck the **"Disabled"** option in the **"Status"** page, click the **"Save"** button, click the **"Apply"** button, it will ask whether to restart the gateway service immediately (see the figure below), click **"Apply"** to take effect immediately, Click **"OK"** again. Then refresh the page, wait for the gateway service to restart and update the interface information, as shown in the following figure.



1.3.2 Hide or Show AP

The hotspot can be hidden only when the AP is on. In the **"Status"** page, check the **"Hidden"** option, click the **"Save"** button and then click **"OK"** , you can also hide or display the WiFi hotspot, as shown in the figure below.



AP CONFIGURATION

2G

☐ Disabled ?

☒ Hidden ?

SSID

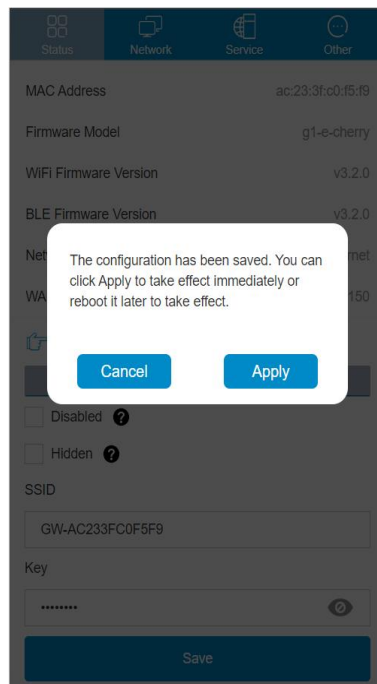
GW-AC233F15AA01

Key

Save

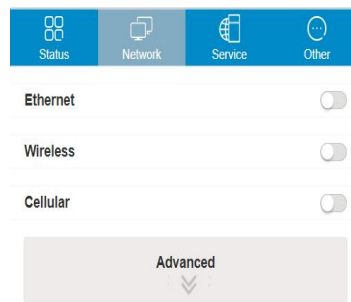
1.3.3 Configuring AP Password

In the AP setting column under **"Status"** of the gateway WEB configuration interface , the user can set the connection password in AP mode (see the figure below). After clicking the **"Apply"** button, it will take effect immediately, or you can click **"Cancel"** to cancel, and it will take effect after restarting.



2. Network Configuration

The G1 Bluetooth gateway can connect to the network in three ways. The default is in the dhcp mode of Ethernet, as shown in the figure below.



Note: Cellular connection is not recommended for the time being.

2.1 Ethernet Connection

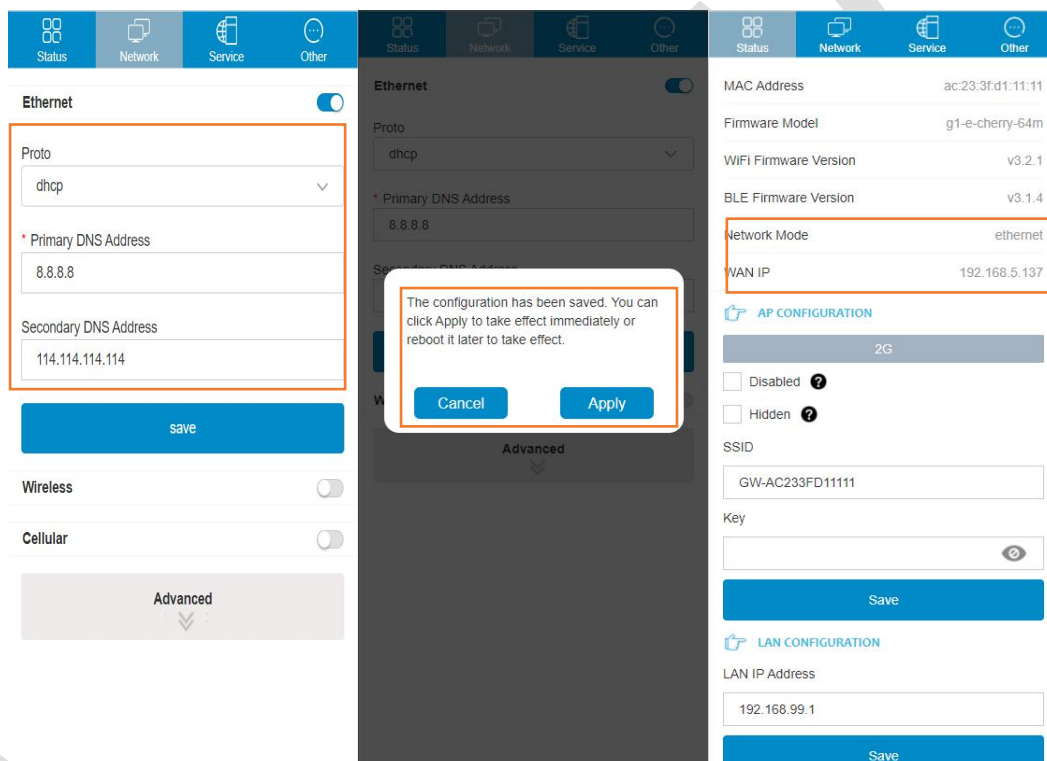
The G1 Bluetooth gateway can be directly connected to the Ethernet to realize data communication with the server.

Ethernet mode has two sub-modes, dhcp mode (default) and static mode. In dhcp mode, the gateway automatically obtains an IP address from the upper-level router, and in static mode, a fixed IP address is manually assigned to the gateway.

2.1.1 DHCP Mode

In the Ethernet setting column of the gateway configuration interface, select the dhcp mode, click the **"Apply"** button, then it will ask whether the configuration has been saved to apply, click **"Apply"** to take effect immediately, you can also click **"Cancel"** , and then manually

Reboot takes effect. The G1 Bluetooth gateway is connected to the network cable of the upper-level router. After configuring the application, it will automatically obtain an IP address from the upper-level router through the DHCP² protocol and connect to the network. See below. In the status item of the gateway WEB configuration interface, the current connection status and the current gateway address will be displayed.



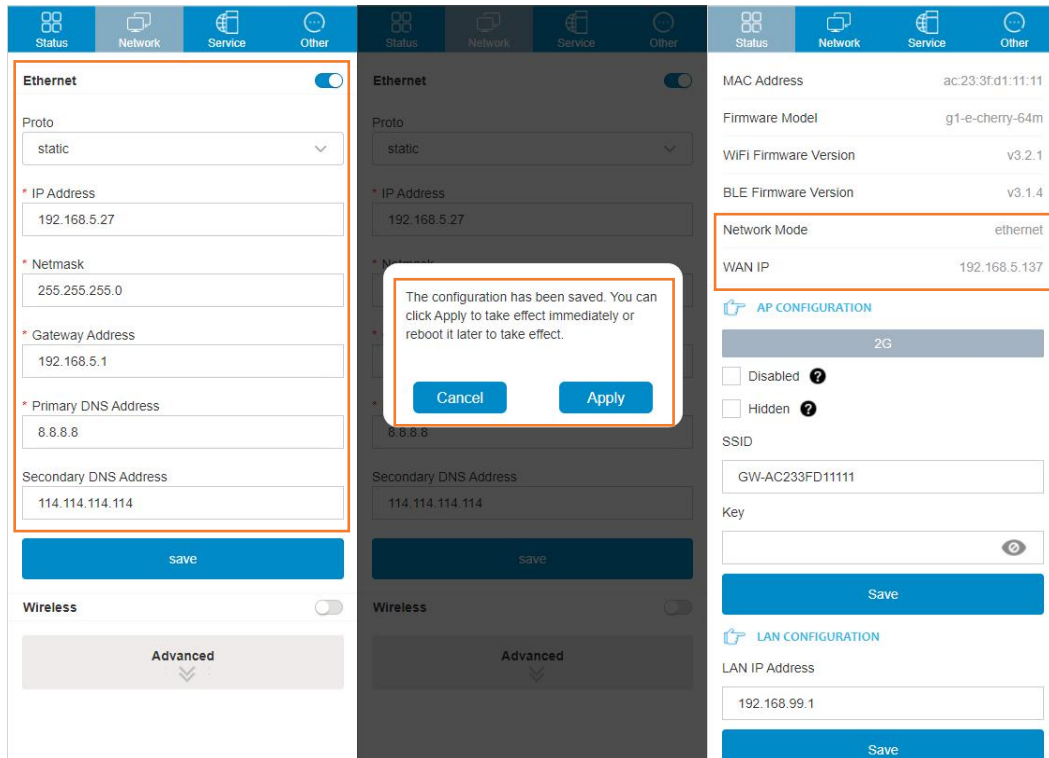
2.1.2 Static Mode

In the Ethernet setting column of the gateway configuration interface, select the static mode, fill in the IP address information, and click the **"Apply"** button. At this time, it will ask whether the configuration has been saved or not. Click **"Apply"** to take effect immediately. You can also click **"Cancel"**, the manual restart will take effect later. The G1 Bluetooth gateway is connected to the network cable of the upper router. After configuring the application gateway, it will

² Dynamic Host Configuration Protocol

automatically connect to the network. See below.

In the status item of the gateway WEB configuration interface, the current connection status and the current gateway address will be displayed.



The interface fields are described in detail below.

field	name	describe
Mode	model	dhcp mode or static mode
IP Address	IP address	Make sure the ip address is not occupied
Net Mask	subnet mask	Confirm that the subnet mask of the upper router is the same
Gateway Address	default gateway	Consistent with the ip address of the upper router
Primary DNS	Preferred DNS	Preferred DNS address, default 8.8.8.8
Secondary DNS	Alternate DNS	Alternate DNS address, default 114.114.114.114

2.2 Wireless connection

The G1 Bluetooth gateway can be used as a WLAN client to connect to a WLAN AP to realize data communication with the server.

Wireless mode has two sub-modes, dhcp mode and static mode. In dhcp mode, the gateway automatically obtains an IP address from the upper-level router, and in static mode, a fixed IP address is manually assigned to the gateway.

2.2.1 DHCP Mode

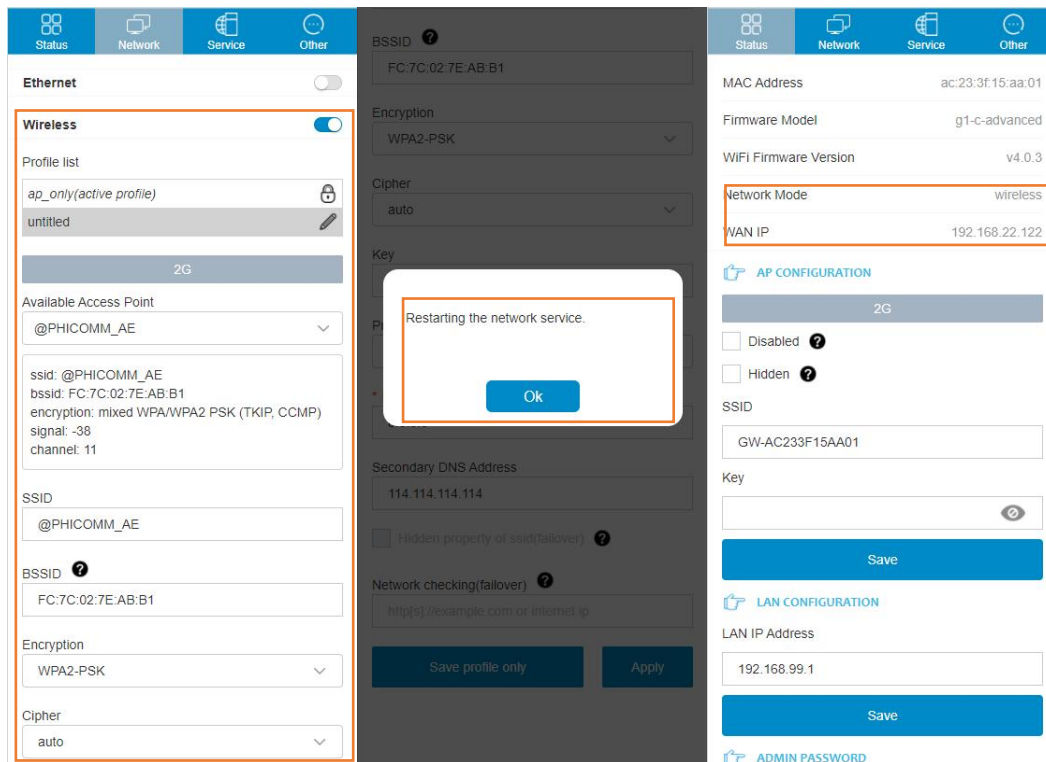
In the Wireless setting column of the Network configuration interface, Profile list is a list of configured WLAN SSIDs.

Select the WLAN SSID to be connected by the gateway from the Available Access Point list, and the corresponding related information, such as ssid, bssid (MAC address), encryption and signal channel, will be displayed in the text box below.

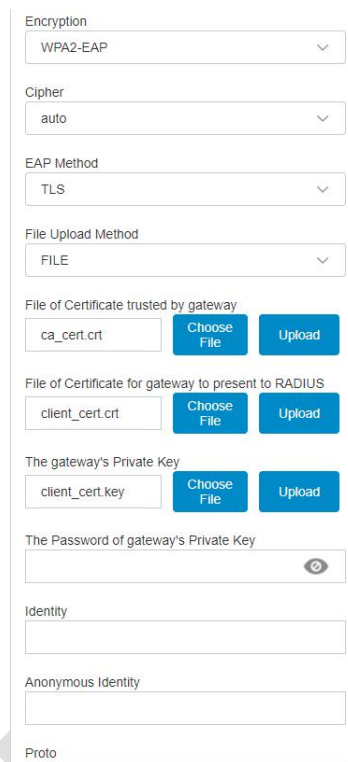
We only need to select the corresponding SSID to fill in the password, select the dhcp mode, click **"Save profile only"** to add only the WiFi profile, and then **"Apply"** to restart the network service.

If it has been connected to the specified WLAN, the IP of the connected WLAN will be displayed in the status item, otherwise N/A will be displayed in the status item.

In order to meet the needs of different customers, we can fill in the hidden WiFi SSID in the SSID column; we can also fill in the router MAC address in the BSSID column to lock it to the corresponding router (the SSID name can be set to the same, and the MAC is unique).



In addition, we have also added support for 802.1x enterprise AP connections. You can select the corresponding encryption method and related configuration in the Encryption column (see the figure below). The specific verification method corresponds to the relevant configuration of the upper-layer router, and the user needs to fill in it by himself.



Encryption
WPA2-EAP

Cipher
auto

EAP Method
TLS

File Upload Method
FILE

File of Certificate trusted by gateway
ca_cert.crt

File of Certificate for gateway to present to RADIUS
client_cert.crt

The gateway's Private Key
client_cert.key

The Password of gateway's Private Key

Identity

Anonymous Identity

Proto

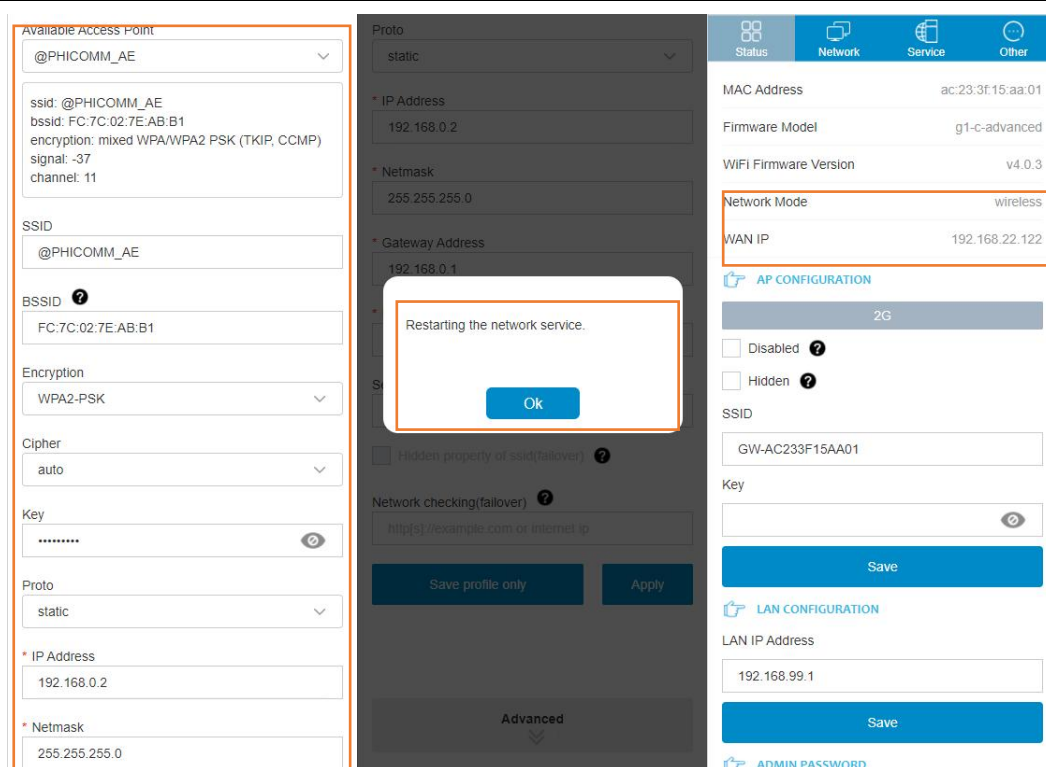
2.2.2 Static Mode

In the Wireless setting column of the Network configuration interface, Profile list is a list of configured WLAN SSIDs.

Select the WLAN SSID to be connected by the gateway from the Available Access Point list, and the corresponding related information, such as ssid, bssid (MAC address), encryption and signal channel, will be displayed in the text box below.

We only need to select the corresponding SSID to fill in the password, select the static mode, fill in the IP address information, click **"Save profile only"** to add only the WiFi profile, and then **"Apply"** to restart the network service.

If it has been connected to the specified WLAN, the IP of the connected WLAN will be displayed in the status item, otherwise N/A will be displayed in the status item.

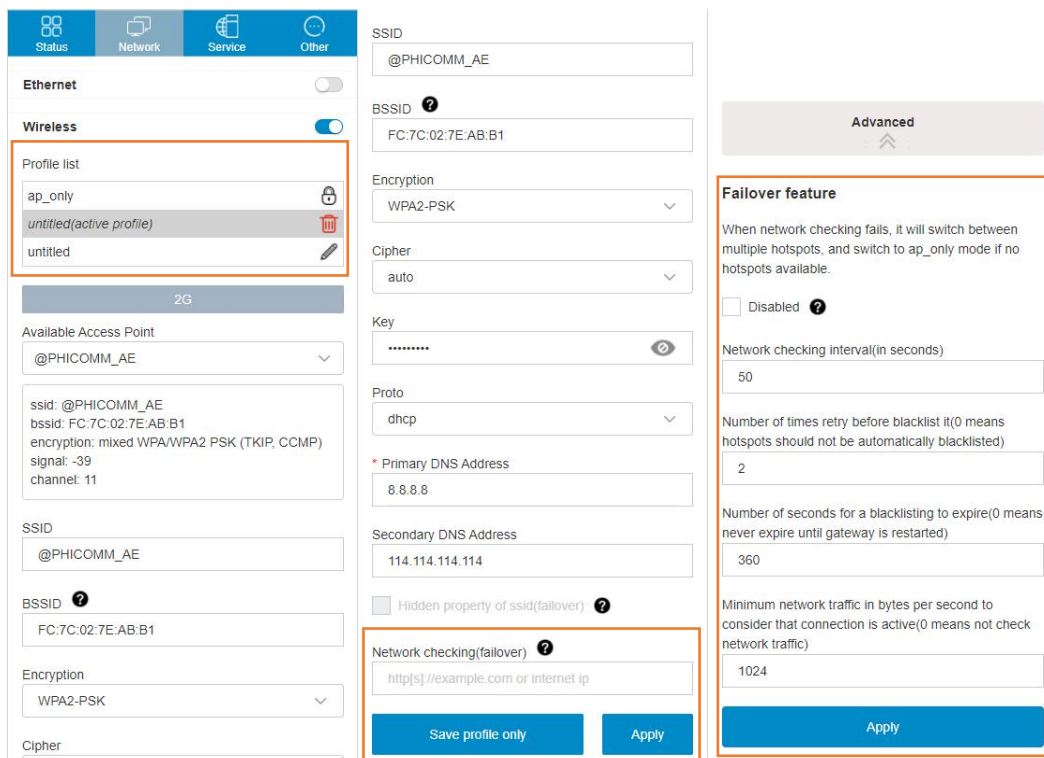


The interface fields are described in detail below.

field	name	describe
Available Access Point	Available WLANs nearby	Automatically obtain available nearby WLAN SSIDs as a client
SSID	selected SSID	The SSID selected by default, if it is hidden, you can modify it by yourself
BSSID	MAC address	The corresponding router can be locked by MAC
Encryption	Encryption	Various encryption methods are available
Cipher	Wireless Security Encryption Protocol	set on the wireless router. It has nothing to do with how to connect to the network, the default is auto
Key	WLAN password	The connection password for connecting to the specified WLAN

Proto	model	dhcp mode or static mode
IP Address	IP address	Make sure the ip address is not occupied
Net Mask	subnet mask	Confirm that the subnet mask of the upper router is the same
Gateway Address	default gateway	Consistent with the ip address of the upper router
Primary DNS	Preferred DNS	Preferred DNS address, default 8.8.8.8
Secondary DNS	Alternate DNS	Alternate DNS address, default 114.114.114.114
Hidden property of ssid(failover)	The hidden attribute corresponding to the SSID	For a more detailed explanation, see WiFi Failover
Network checking(failover)	IP address or domain name to use for failover network detection	For a more detailed explanation, see WiFi Failover

2.2.3 WiFi Failover



The screenshot displays the MINEW configuration interface with the following sections and settings:

- Top Navigation:** Status, Network, Service, Other.
- Ethernet:** Toggle switch is off.
- Wireless:** Toggle switch is on.
- Profile list:**
 - ap_only (with lock icon)
 - untitled(active profile) (with trash icon)
 - untitled (with edit icon)
- Available Access Point:** @PHICOMM_AE
- Network Details:**
 - ssid: @PHICOMM_AE
 - bssid: FC:7C:02:7E:AB:B1
 - encryption: mixed WPA/WPA2 PSK (TKIP, CCMP)
 - signal: -39
 - channel: 11
- Advanced Settings:**
 - SSID: @PHICOMM_AE
 - BSSID: FC:7C:02:7E:AB:B1
 - Encryption: WPA2-PSK
 - Cipher: auto
 - Key: [masked]
 - Proto: dhcp
 - * Primary DNS Address: 8.8.8.8
 - Secondary DNS Address: 114.114.114.114
 - Hidden property of ssid(failover): [unchecked]
 - Network checking(failover): [http\[s\]://example.com](http[s]://example.com) or internet ip
 - Buttons: Save profile only, Apply
- Failover feature (Expanded):**
 - Description: When network checking fails, it will switch between multiple hotspots, and switch to ap_only mode if no hotspots available.
 - Enabled: [checked]
 - Network checking interval(in seconds): 50
 - Number of times retry before blacklist it(0 means hotspots should not be automatically blacklisted): 2
 - Number of seconds for a blacklisting to expire(0 means never expire until gateway is restarted): 360
 - Minimum network traffic in bytes per second to consider that connection is active(0 means not check network traffic): 1024
 - Button: Apply

Note:  there are detailed explanations

After the failover function is enabled, once the network check of the hotspots in the Profile List fails, the gateway will switch the hotspots in the Profile List in sequence. When a hotspot is blacklisted, the gateway will not retry until the blacklist expires. When the hotspots in the Profile List are unavailable, the gateway will switch to AP_ONLY mode. During the AP_ONLY mode, if the hotspots in the Profile List are found to be back to normal, the gateway will automatically retry to connect to the hotspots in the Profile List.

AP_ONLY Mode: AP_ONLY mode means that the gateway does not connect to the upper-level WiFi hotspot, but only transmits its own hotspot, so that after the upper-level WiFi hotspot fails, the user can use the gateway's own hotspot to re-enter the configuration page.

specific properties

field	describe
Hidden property of ssid	the WiFi ssid corresponding to the current profile is hidden . (When the failover function is enabled, do not arbitrarily change the hidden properties of the hotspot, otherwise the failover function may not find the hotspot.)
Network checking	whether the current profile network is normal by testing the specified URL or IP . When failover is enabled, the gateway will verify the connection reliability of this hotspot by pinging this IP (make sure ICMP protocol is turned on) or wget URL (make sure there is a page). (If it is not set, it will only detect whether WiFi can be connected, not whether it can access the external network)

Common properties

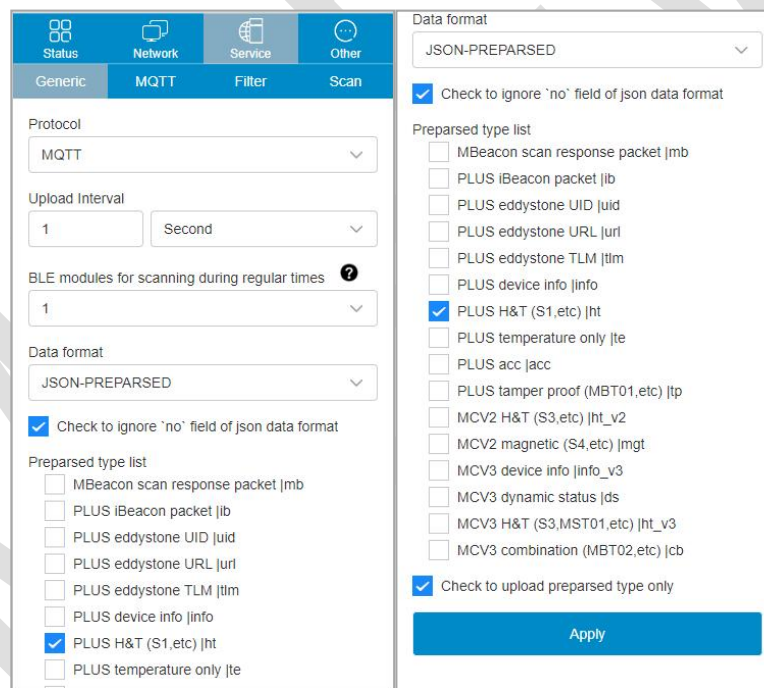
field	describe
Enable failover	Whether to enable failover (it can solve the problem that the gateway cannot send its own AP when the upper-level hotspot is unavailable)
Network checking interval	Network check interval (in seconds)
Number of times retry before blacklist it	The number of times to retry before blacklisting it (0 means the hotspot will not be automatically blacklisted)
Number of seconds for a blacklisting to expire	The number of seconds for the blacklist to expire (0 means it will not expire until the gateway is restarted)
Minimum network traffic in bytes per second to consider that connection is ative	The minimum network traffic (in bytes/second) when the connection is active. If the current network traffic reaches the set value, Network checking will not be performed to detect the current network. When set to 0 it will never be detected, it is recommended not to set to 0.

3. Data Collection & Service Access

3.1 Data Collection

After the G1 Bluetooth gateway is started, it will continue to collect Bluetooth broadcast data, and when the network is available, it will send data to the connected server at a frequency of once a second by default.

The data uploaded by the G1 Bluetooth gateway has three formats (additional formats may be added with firmware updates, subject to actual display), MINEW-CONNECT, JSON-RAW, and JSON-PREPARED. The gateway defaults to using the MINEW-CONNECT format to upload data to the server.



The screenshot shows the configuration interface for the MINEW G1 Bluetooth gateway. The interface is divided into several sections:

- Protocol:** A dropdown menu set to "MQTT".
- Upload Interval:** A dropdown menu set to "1" with a unit of "Second".
- BLE modules for scanning during regular times:** A dropdown menu set to "1".
- Data format:** A dropdown menu set to "JSON-PREPARED".
- Check to ignore 'no' field of json data format:** A checked checkbox.
- Prepared type list:** A list of data types with checkboxes:
 - ☐ MBeacon scan response packet |mb
 - ☐ PLUS iBeacon packet |ib
 - ☐ PLUS eddystone UID |uid
 - ☐ PLUS eddystone URL |url
 - ☐ PLUS eddystone TLM |tlim
 - ☐ PLUS device info |info
 - ☒ PLUS H&T (S1,etc) |ht
 - ☐ PLUS temperature only |te
 - ☐ PLUS acc |acc
 - ☐ PLUS tamper proof (MBT01,etc) |tp
 - ☐ MCV2 H&T (S3,etc) |ht_v2
 - ☐ MCV2 magnetic (S4,etc) |mgt
 - ☐ MCV3 device info |info_v3
 - ☐ MCV3 dynamic status |ds
 - ☐ MCV3 H&T (S3,MST01,etc) |ht_v3
 - ☐ MCV3 combination (MBT02,etc) |cb
- Check to upload prepared type only:** A checked checkbox.
- Apply:** A blue button at the bottom.

field	name	describe
Upload Interval	Time interval for uploading broadcast packets	The unit can be selected, the default is 1s
BLE modules for scanning during regular	/	By default, only one Bluetooth module is used for scanning. If multiple modules are set for simultaneous scanning and the platform issues a BLE

times		connection command, the gateway will automatically switch to scanning with one Bluetooth module. When there is no BLE connection command, the gateway will automatically switch back to scanning with multiple Bluetooth modules.
Data format	BLE advertising packets format	The data uploaded by the gateway has three formats: MINEW-CONNECT, JSON-RAW, and JSON-PREPARED, defaults to MINEW-CONNECT
Check to ignore `no` filed of json data format	/	The default value is True, which means that the no field is ignored. The no field indicates which Bluetooth module receives the broadcast packet, and is generally not required for applications.
Prepared type list	The pre parsed ble types that supports	Only available to JSON-PREPARED. See <i>minew-connect-v3@gateway-interface.pdf</i> for more details.
Check to upload prepared type only	Upload only the options checked in the Prepared type list	The default value is True. If you want to parse BLE broadcast packets by yourself, you can uncheck it, gateway will upload the raw data to platform that you can use.

Note: Due to the length of this part, please refer to the *minew-connect-v3@gateway-interface.pdf* document for detailed data formats.

3.2 Service Access

The G1 Bluetooth gateway currently supports the use of MQTT, HTTP, and TCP network protocols to communicate with servers locally or on the Internet.

Users can choose a suitable network protocol according to their server architecture, and the MQTT protocol is recommended.

Service access of the G1 Bluetooth gateway requires back-end MQTT or HTTP or TCP service support. The G1 Bluetooth gateway is connected to the TagCloud IoT service in Yunliwu by default when it leaves the factory.

The G1 Bluetooth gateway is equipped with the TagCloud IoT service and console in Yunliwuli

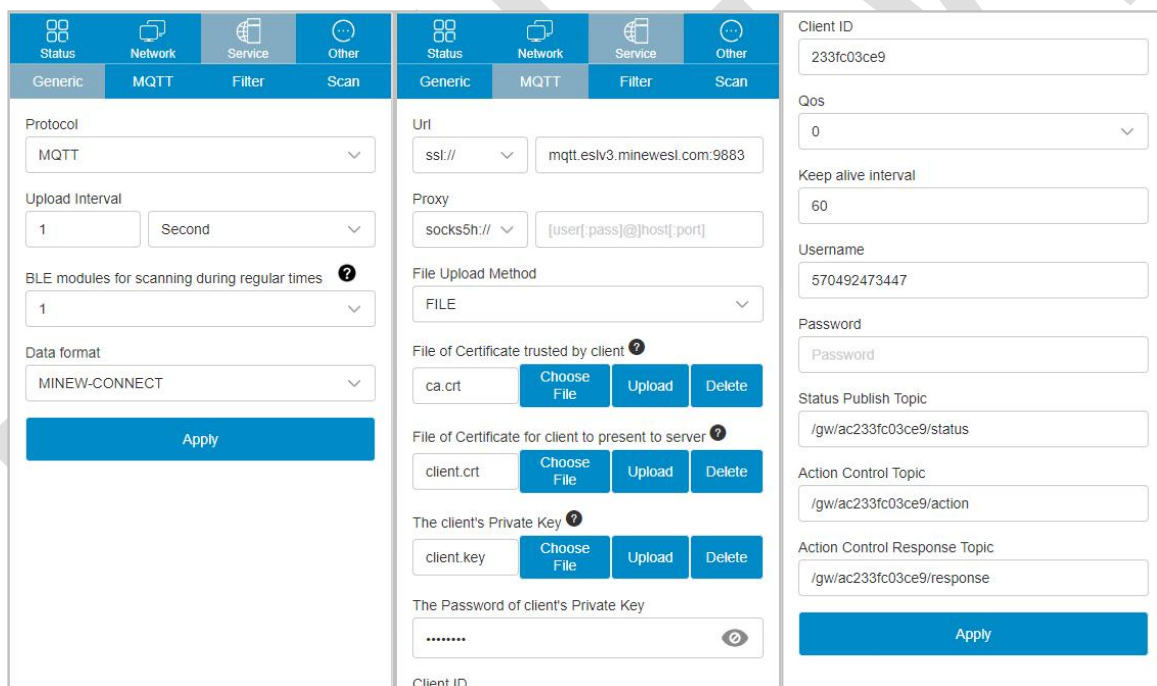
by default as a demonstration, you can quickly experience the application of the Bluetooth gateway.

3.2.1 MQTT Access

When using MQTT access, the G1 Bluetooth gateway supports the function of regularly uploading Bluetooth data and remote command control.

The G1 Bluetooth gateway can support the MQTT 3.1.1 protocol standard server. For the specific protocol, please refer to the MQTT 3.1.1 protocol document, and see <http://mqtt.org for details>.

For MQTT related libraries, you can check <https://github.com/mqtt/mqtt.github.io/wiki/libraries>.



field	name	describe
Protocol	Service access method	Choose MQTT
Url	Url of MQTT server	scheme+domain name/IP Address+port scheme: optional values tcp://, ssl://

		port: When not filled in, when the scheme is tcp://, the port defaults to 1883, and when it is ssl://, the port defaults to 8883 Default: tcp://tagcloud.minew.com:2883
proxy	proxy server	Setting up a sock5 proxy server for MQTT
Client ID	MQTT Client ID	The Client ID defined by the MQTT protocol is the identity of the client Default: \${gatewayMAC} \${gatewayMAC} is the hexadecimal lowercase character form of the gateway's Mac, such as aabbcddeeff
Qos	MQTT QoS level	The QoS level defined by the MQTT protocol. QoS=0: at most once, possibly duplicate or lost. QoS=1: At least once, possibly repeated. QoS=2: Yes and only once. Default: 0
keep alive interval	Heartbeat interval for MQTT	The keep alive interval defined by the MQTT protocol defaults to 10 seconds. Some MQTT cloud servers do not allow such a small heartbeat interval. For example, Alibaba Cloud needs to adjust this parameter to 60 seconds to connect.
Username	MQTT username	Username defined by the MQTT protocol
Password	MQTT password	Ciphers defined by the MQTT protocol
Status Publish Topic	Topic for collecting data and publishing	Topic for the gateway to publish BLE data Default: /gw/\${gatewayMac}/status \${gatewayMac} is the hexadecimal lowercase character form of the gateway's Mac, such as /gw/aabbcddeeff/status.
Action Control Topic	Topic of control command message	It is used by the gateway to receive control commands. The message is in JSON format. Currently, only the "restart" command is supported. Default: /gw/\${gatewayMac}/action \${gatewayMac} is the hexadecimal lowercase character form of the gateway's Mac, such as /gw/aabbcddeeff/action.
Action Control Response Topic	Topic that controls command response	It is used by the gateway to respond to control commands, and the message is in JSON format. Default: /gw/\${gatewayMac}/response. \${gatewayMac} is the hexadecimal lowercase character form of the gateway's Mac, such as

/gw/aabbcddeeff/response.

MQTT with Proxy Method

When using MQTT access, it supports Proxy Socks5 (the proxy resolves the URL hostname).

Any of the following formats will work:

username: password@host:1080

username:password@host

username@host:1080

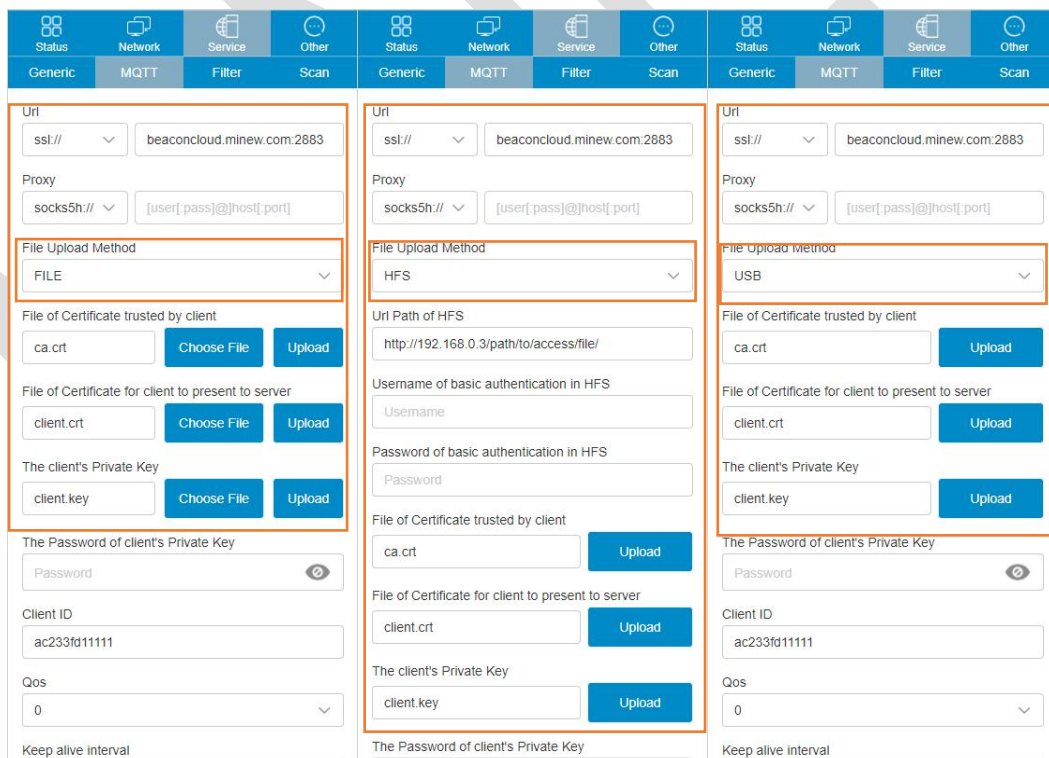
username@host

host:1080

host

MQTT With SSL/TLS Method

When using MQTT access, it supports SSL/TLS protocol. After selecting "ssl://" in the drop-down list of Url parameters, the parameters displayed are used to configure the loading certificate to the gateway. Both USB and HFS are supported. As shown below.



The image displays three side-by-side screenshots of the MQTT configuration interface, each showing a different 'File Upload Method' selected in the 'File Upload Method' dropdown menu. The interface is divided into several sections for configuration.

- Uri:** A dropdown menu set to 'ssl://' and a text field containing 'beaconcloud.minew.com:2883'.
- Proxy:** A dropdown menu set to 'socks5h://' and a text field containing '[user[:pass]]@[host[:port]]'.
- File Upload Method:** A dropdown menu with three options: 'FILE', 'HFS', and 'USB'.
- File of Certificate trusted by client:** A text field containing 'ca.crt' and a blue 'Upload' button.
- File of Certificate for client to present to server:** A text field containing 'client.crt' and a blue 'Upload' button.
- The client's Private Key:** A text field containing 'client.key' and a blue 'Upload' button.
- The Password of client's Private Key:** A text field with a password icon and a blue 'Upload' button.
- Client ID:** A text field containing 'ac233fd11111'.
- Qos:** A dropdown menu set to '0'.
- Keep alive interval:** A text field.

The three screenshots show the configuration for 'FILE', 'HFS', and 'USB' methods respectively. The 'HFS' method includes an additional 'Uri Path of HFS' field set to 'http://192.168.0.3/path/to/access/file/'.

field	name	describe
File Upload Method	How to upload the certificate	Optional values: USB and HFS (Http File Server) USB: Upload the certificate using USB HFS: Upload certificates using HFS FILE: Upload certificate via browser
Url Path of HFS	URL PATH part of HFS	Support HFS for gateway to download certificate file Default: http://192.168.0.3/ssl/
Username of basic authentication in HFS	Username for HFS Basic Authentication	If HFS does not have basic authentication, you can leave this parameter blank
Password of basic authentication in HFS	Password for HFS Basic Authentication	If HFS does not have basic authentication, you can leave this parameter blank
File of Certificate trusted by client	Gateway trusted CA certificate	File name of CA certificate in U disk/HFS USB: The gateway will directly download the certificate file to the root directory of the U disk HFS: The gateway will download the certificate file directly to the \${Urlpath} path FILE: The gateway will download the certificate file through the browser
File of Certificate for client to present to server	Gateway's certificate	The file name of the gateway certificate in the U disk/HFS USB: The gateway will directly download the certificate file to the root directory of the U disk HFS: The gateway will download the certificate file directly to the \${Urlpath} path FILE: The gateway will download the

		certificate file through the browser
The client 's Private Key	The private key of the gateway certificate	filename of keyfile in HFS USB: The gateway will directly download the certificate file to the root directory of the U disk HFS: The gateway will download the certificate file directly to the \${Urlpath} path FILE: The gateway will download the certificate file through the browser
The Password of client 's Private Key	Password for the private key of the gateway certificate	If the private key of the gateway certificate is encrypted, this password must be provided

The configuration instructions for SSL/TLS certificates are as follows:

USB Upload Method

method of uploading the certificate file using USB is as follows:

1. "Upload Way" on the web configuration page ;
2. Name the certificate to be uploaded as the file name of the corresponding input box;
3. Copy the named certificate file to the root directory of the U disk;
4. Insert the U disk into the USB port of the gateway;
5. Click the corresponding Upload button on the configuration page to upload the corresponding certificate file.

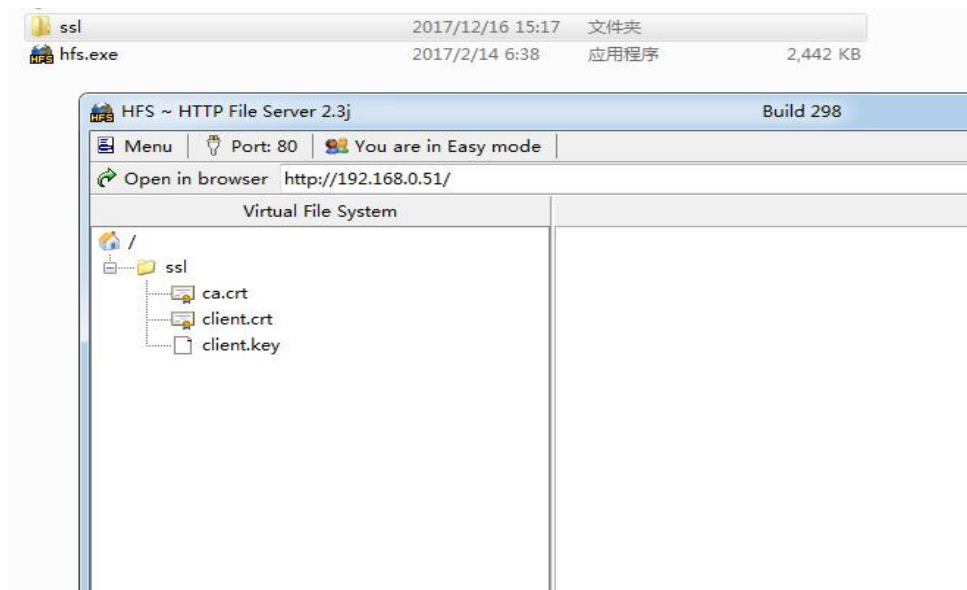
HFS Upload Method

The method of uploading the certificate file in HFS mode is as follows:

Preliminary preparation: link under HFS software: <https://sourceforge.net/projects/hfs/>

Click to open the hfs.exe software to open the HFS service. The premise is that the gateway must be configured with a network to communicate with HFS.

1. " Upload Way" on the webpage configuration page ;
2. Name the certificate to be uploaded as the file name of the corresponding input box;
3. Add the named certificate file to the path of HFS, as shown in the following figure;



4. Fill in the path where the certificate is located in the Urlpath input box, for example: <http://192.168.0.51/ssl> ;
5. Click the corresponding Upload button on the configuration page to upload the corresponding certificate file.

FILE Upload Method

The method of uploading the certificate file in FILE mode is as follows:

1. "File Upload Method" on the web configuration page ;
2. "Choose File" on the webpage configuration page to select the corresponding certificate file;
3. Click the corresponding "Upload" button on the configuration page to upload the corresponding certificate file.

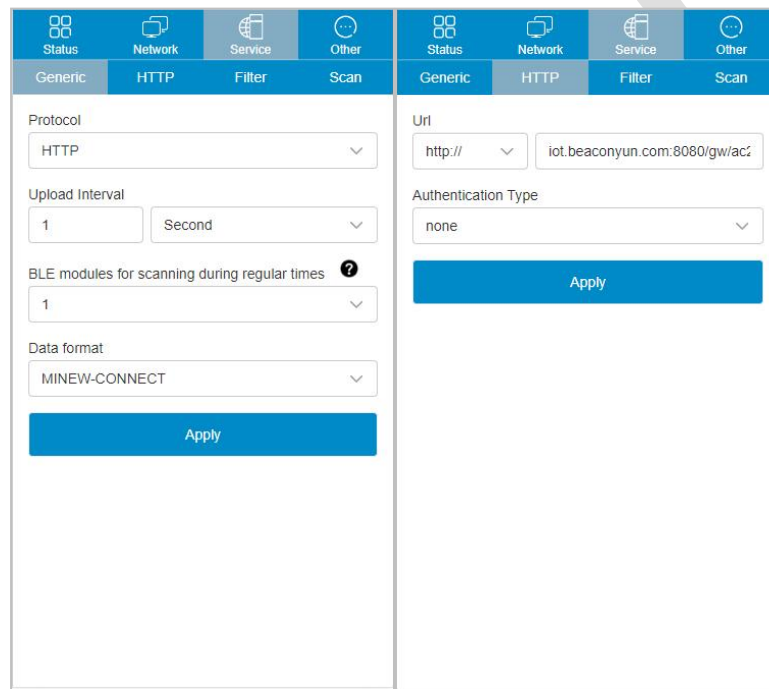
3.2.2 HTTP Access

When using HTTP access, the G1 Bluetooth gateway only supports the function of regularly uploading and collecting data.

G1 Bluetooth gateway can support HTTP/HTTPS protocol web server. In the service connection setting column of the gateway WEB configuration interface, select HTTP, and fill in the URL information of the HTTP server, and it will take effect after saving.

HTTP server development instructions:

1. The HTTP server must be able to receive requests for the HTTP Post method.
2. The HTTP server should receive any packets uploaded by the gateway, including empty packets (this is the request packet used to test if the HTTP server is online). Only after the HTTP server returns HTTP status code 200, the gateway can continue to send the next packet.



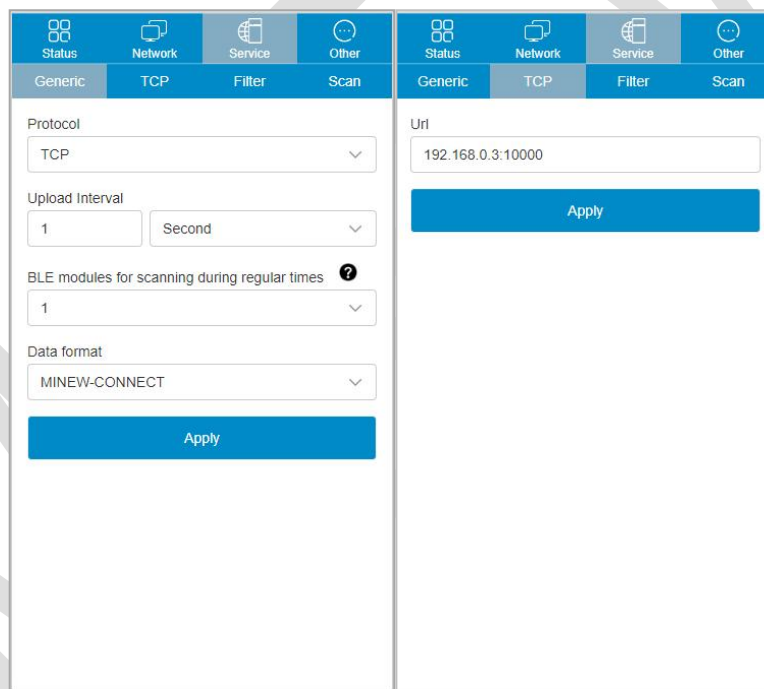
field	name	describe
Protocol	Service access method	select HTTP
Url	URL for collecting data	It is used for the gateway to upload data regularly, starting with http:// or https://, using the POST method http://tagcloud.minew.com:8080/gw/\${gatewayMac}/status \${gatewayMac} is the hexadecimal lowercase character of the gateway's Mac, for example http://tagcloud.minew.com:8080/gw/aabbcddeeff/status.
Authentication Type	HTTP Authentication Type	Optional values: none (do not use HTTP authentication), basic (use HTTP basic Authentication)

		Default: none
Username	username	Username for HTTP basic authentication
Password	password	Password for HTTP basic authentication

3.2.3 TCP Access

When using TCP access, the G1 Bluetooth gateway only supports the function of regularly uploading and collecting data.

G1 Bluetooth gateway can support the web server of TCP protocol. In the service connection setting column of the gateway WEB configuration interface, select TCP, and fill in the URL information of the TCP server, and it will take effect after saving.

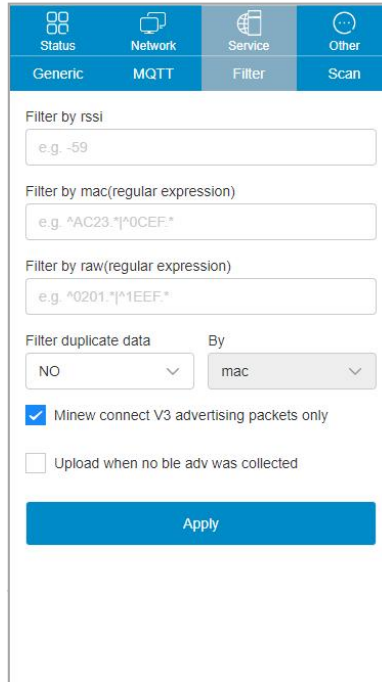


The screenshot shows the 'Service' tab of the gateway configuration interface. Under the 'TCP' sub-tab, the 'Protocol' is set to 'TCP'. The 'Upload Interval' is set to '1' second. The 'BLE modules for scanning during regular times' is set to '1'. The 'Data format' is set to 'MINEW-CONNECT'. An 'Apply' button is visible at the bottom of the settings. On the right side, the 'Url' field is set to '192.168.0.3:10000' and another 'Apply' button is present.

field	name	describe
Protocol	Network protocol	select TCP
Url	URL for collecting data	It is used for the gateway to upload data regularly, The URL format is {host/IP}:{port} For example 192.168.0.3:10000

3.3 Data Filtering

The data filtering for access to the three services is the same, as shown in the following figure.



field	name	describe
Filter by rssi	Rssi value filtering	Only upload data whose signal strength is greater than this value, in dBm When not filled, it means no filtering, the value can be filled in $-100 < rssi < 0$ For example: when filling in -67, the weak signal of -68 and below will not be uploaded to the server
Filter by mac	Filter BLE mac by regex	Regular expression for BLE mac For example: only want to receive BLE mac data starting with AC23 or 0CEF, the regular expression is: <code>^AC23.* ^0CEF.*</code> Note: Here BLE mac is not case sensitive, no colon.
Filter by raw	Filter raw data by regular expression	Regular expression for raw data This filter is applicable to all data formats. This filter condition can be used as long as the broadcast packets of the BLE device are regular. For example: only want to receive UUID as <code>FDA50693A4E24FB1AFCFC6EB07647825</code> the iBeacon broadcast package, the expression can

		be written as ^.* FDA50693A4E24FB1AFCFC6EB07647825 .* Note: Although this version does not have the Filter by blename option, it is possible to use Filter by raw instead of Filter by blename by converting the ascii code of the BLE name into hexadecimal form.
Filter duplicate data	Whether to filter duplicate data within the upload interval	Default NO, optional YES, NO; There are three ways to set filtering conditions: mac: Broadcast packets with the same mac address are considered duplicate data; mac+type: broadcast packets with the same mac address and the same type are considered duplicate data; mac+raw: broadcast packets with the same mac address and the same raw data are considered duplicate data
Minew Connect V3 advertising packets only	/	By default, only broadcast packets using the Minew Connect V3 protocol are received. <i>Please uncheck to receive more BLE advertising packets.</i>
Upload when no ble adv was collected	Should an empty packet be uploaded when no BLE broadcast is scanned	To save data usage or cost, it is not uploaded by default. If you want to use it to determine if the gateway is online, you can enable this option. In this way, the gateway will upload data at the interval of the <i>upload interval</i> .

3.4 Scan Parameter Configuration

This part of the parameters is used to configure the scanning strategy of the Bluetooth module. As shown below.

Status

Network

Service

Other

Generic

MQTT

Filter

Scan

☐ Minew only
☐ Active scan

Apply

filed	name	describe
Minew only	Whether to only scan Minew devices	1. When enabled, only data that conforms to the Minew advertising format will be scanned, which is related to the Minew Connect V3 advertising format. 2. The difference between this option and the "Minew Connect V3 advertising packets only" option is that "Minew Only" is used by the BLE firmware, while the "Minew Connect V3 advertising packets only" option is used by the WiFi firmware. 3. <i>Please uncheck to receive more BLE advertising packets.</i>
Active Scan	Whether to enable active scanning	1. There are two types of broadcast packets of BLE tags: broadcast packets and scan reply packets. The broadcast packet is actively broadcast by the BLE tag; the scan reply packet is passively broadcast after the BLE tag receives the request after the scanner sends a request. 2. The scanning strategy of the BLE module of the gateway is correspondingly divided into two types, active scanning and passive scanning. Active scanning will send

a request for reply packets to the BLE tag, while passive scanning will only receive broadcast packets and will not send a request for reply packets.

3. If your BLE tag does not have reply packets, you can uncheck this option, which can improve the scanning rate of BLE broadcast packets.

3.5 Remote Control (Only MQTT)

Remote control allows commands to be sent to the gateway, and it can also pass command data to BLE devices through the gateway.

Please refer to the *minew-connect-v3@gateway-interface.pdf* document for more information.

3.6 Disconnected Storage

The gateway off-network storage function is not available yet in the grapes firmware.

4. Other Configuration

4.1 Top Lights Settings

The meanings of the top lights are as follows:

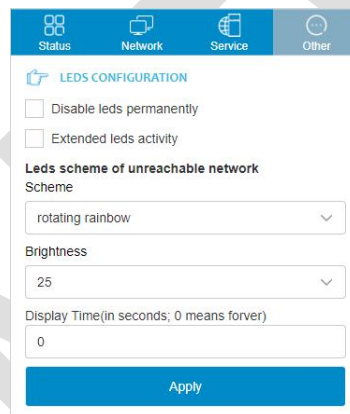
state	meaning
Static six-color	Indicates that the gateway is starting up, and pressing the RESET button in

light	this state is invalid.
Dynamic white light rotation	The gateway has been started and is trying to connect to the server
Dynamic six-color light rotation (can be configured as a single color of red, yellow, and white)	If the gateway fails to connect to the server, it means that the network configuration of the gateway is unsuccessful or the server is not online.
Dynamic breathing light	The gateway has successfully connected to the server (Note: After the gateway is in the dynamic breathing light by default, the lights will be turned off after a one-minute delay to enter the energy-saving mode. Local tyrants can set the long-light mode on the configuration page.)
lights off	Indicates that the gateway is not powered on or is in power-saving mode

By default, the Bluetooth gateway is in the normal working state (breathing light state), and the lights will be turned off automatically after a delay of about 1 minute. You can configure the options of the lights in LEDS CONFIGURATION, as shown in the figure below .

field	describe
Disable leds permanently	YES: Turn off the lights, only turn on when the gateway system is turned on, turn off the lights after the system is fully started NO: turn on the lights
Extended leds activity	YES: The lantern will be on for a long time NO: The colored lights are turned off after a delay of 1 minute, but if the gateway cannot connect to the server, the six-color lights will always turn on and rotate.

	Note: When Disable leds permanetnly is YES, this field is invalid
Scheme	When the gateway is not connected to the server, the available color light schemes are: six-color light rotation, pure red, pure yellow, pure white
Brightness	brightness value of the colored lights is set to 1-25 , and the value of 25 is the brightest.
Display Time (in seconds, 0 means forever)	The setting of the lighting time value, calculated in units of seconds, 0 means it will be on for a long time



4.2 Time Configuration

The gateway does not have an RTC clock, so it is necessary to synchronize the time of the NTP server to obtain the accurate time or manually synchronize the time of the host where the browser is located to the gateway.

4.2.1 Synchronization Using NTP Server

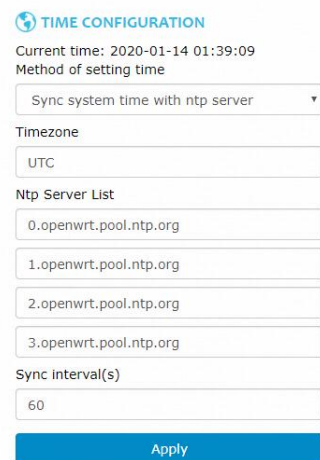
Both the time zone and the NTP server address can be set. In the time configuration column of "Other" on the web page, you can set the time zone (UTC by default) and NTP server (default is the ntp server of openwrt). After filling in your own information, click the "Apply" button to

complete the configuration. Effective immediately. As shown below.

The default UTC time zone is 0 time zone. If the timezone is in the east time zone, it can be set to UTC-time zone number. For example, if it is set to UTC-8, it is the time zone of our east eighth zone;

If the timezone can be set in the western time zone, it can be set to UTC+time zone number. For example, if it is set to UTC+8, it is the time zone of the western eighth zone.

Sync interval Indicates the interval for synchronizing time with the server. For example, if it is set to 60, it means that the time synchronization with the server will be performed automatically every minute.



The image shows a 'TIME CONFIGURATION' interface. At the top, it displays 'Current time: 2020-01-14 01:39:09'. Below this is a section 'Method of setting time' with a dropdown menu set to 'Sync system time with ntp server'. The 'Timezone' is set to 'UTC'. Under 'Ntp Server List', there are four input fields, each containing '0.openwrt.pool.ntp.org', '1.openwrt.pool.ntp.org', '2.openwrt.pool.ntp.org', and '3.openwrt.pool.ntp.org'. The 'Sync interval(s)' is set to '60'. At the bottom, there is a blue 'Apply' button.

Reminder: China time zone can also be set to CST-8. For more detailed Timezone settings, please check <https://wiki.openwrt.org/doc/uci/system>

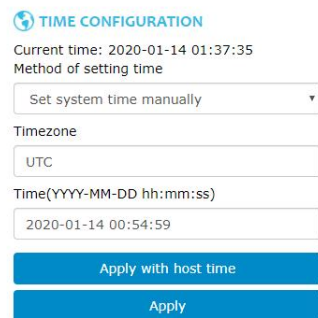
4.2.2 Manual Time Setting

To manually set the gateway system time, you can click Apply with host time or click Apply. Current time shows the time of the gateway system. As shown below.

When you click Apply with host time, the gateway will actively obtain the time of the host where

the browser is located and synchronize the time to the gateway. Before clicking Apply with host time, the Timezone parameter must be the same time zone as the time zone of the host where the browser is located, otherwise the time set will be different from the time zone.

When you click Apply, the gateway synchronizes the time at the time obtained to the gateway to the gateway.



TIME CONFIGURATION

Current time: 2020-01-14 01:37:35
Method of setting time

Set system time manually

Timezone

UTC

Time(YYYY-MM-DD hh:mm:ss)

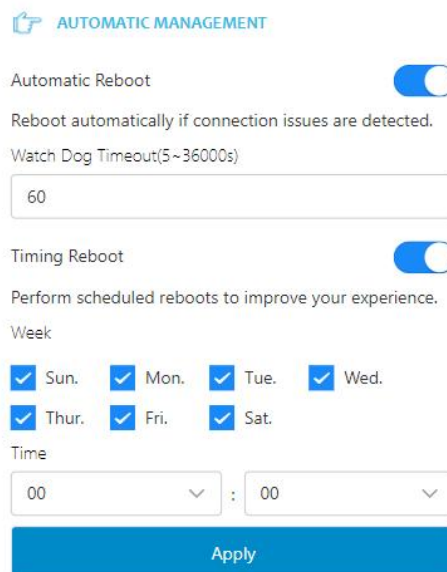
2020-01-14 00:54:59

Apply with host time

Apply

4.3 Automatic Management

The automatic management module is divided into Automatic Reboot (automatic restart) and Timing Reboot (timed restart). As shown below.



AUTOMATIC MANAGEMENT

Automatic Reboot ☒

Reboot automatically if connection issues are detected.

Watch Dog Timeout(5~36000s)

60

Timing Reboot ☒

Perform scheduled reboots to improve your experience.

Week

☒ Sun. ☒ Mon. ☒ Tue. ☒ Wed.
☒ Thur. ☒ Fri. ☒ Sat.

Time

00 : 00

Apply

4.3.1 Automatic Restart

The gateway does not enable the automatic restart function by default. If the gateway is disconnected from the server for more than a specified time during the stable connection with the server (the reason for the disconnection may be the server or the gateway network), the gateway will automatically restart. **After the gateway is restarted, if the server has not been connected to the server, the gateway will not restart again** . The automatic restart will only occur during the process of being disconnected in the middle.

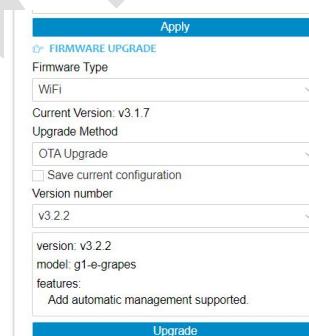
4.3.2 Scheduled Restart

The timing restart function is disabled by default on the gateway. After enabling this function, the gateway will restart at the specified time (the time zone used by the time is the time zone in the previous time configuration).

4.4 Firmware Upgrade

The gateway consists of a WiFi module and a BLE module, so the gateway upgrade involves two types of firmware: WiFi firmware and BLE firmware. **It is recommended to upgrade the BLE firmware first and then the WiFi firmware during the upgrade process** .

Different firmware versions correspond to different function adjustments and upgrades, which can be selected as needed. For details, see the following figure:



The screenshot shows a web-based interface for firmware upgrade. At the top is a blue 'Apply' button. Below it is a section titled 'FIRMWARE UPGRADE' with a blue icon. The 'Firmware Type' is set to 'WiFi'. The 'Current Version' is 'v3.1.7'. The 'Upgrade Method' is set to 'OTA Upgrade'. There is a checkbox for 'Save current configuration' which is unchecked. The 'Version number' is set to 'v3.2.2'. Below this, a box displays the new version details: 'version: v3.2.2', 'model: g1-e-grapes', and 'features: Add automatic management supported.'. At the bottom is a blue 'Upgrade' button.

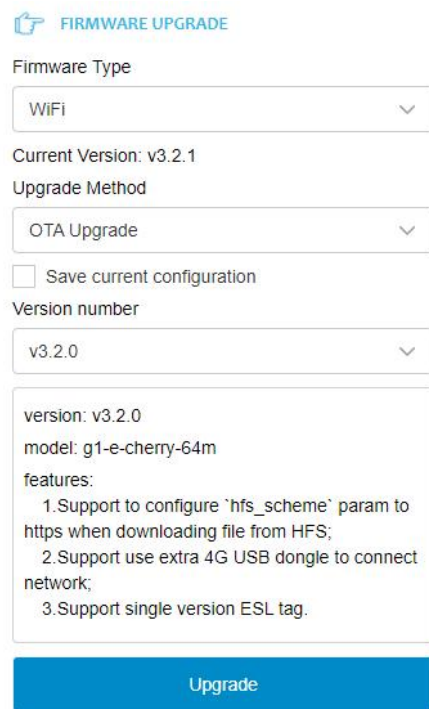
4.4.1 Wifi Firmware Upgrade

WiFi firmware upgrade supports USB upgrade, FILE upgrade, over-the-air upgrade, LAN upgrade and upgrade of custom firmware (essentially over-the-air upgrade).

Note: During the upgrade process, the lights on the top of the gateway will change from off to static six-color lights until they become dynamic lights (the entire upgrade process will last 2 to 3 minutes, USB upgrade is the fastest, and the air upgrade time depends on Network status), **do not disconnect the power of the gateway before the gateway lights become dynamic lights, otherwise there is a risk of bricking** . If the gateway does not display a static six-color light within 2 minutes after clicking Upgrade, it means that the gateway has timed out to obtain the upgrade firmware and the upgrade failed.

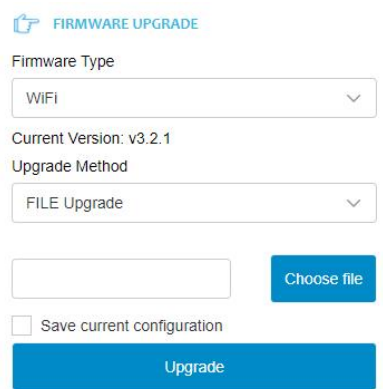
4.4.1.1 Over-the-air Upgrade

In the case that the gateway can connect to the external network, in the Other setting column of the web interface, select WiFi for Firmware Type, and OTA Upgrade for Upgrade Type. If you want to save the current configuration parameters, please check "Save current configuration" and click "**Upgrade**" button and then click "**OK**" to confirm the upgrade. After confirming the upgrade, the firmware will be downloaded for a period of time. If the lights on the top of the gateway are off, it means that the gateway has finished downloading the firmware and is upgrading. Do not disconnect the power supply during the upgrade, wait for about 3 minutes, until the dynamic lights on the top of the gateway light up again, it means that the gateway firmware has been upgraded successfully, and the web interface is shown in the lower bound diagram.



4.4.1.2 FILE Upgrade

In the Other setting column of the web interface, select WiFi for Firmware Type and FILE Upgrade for Upgrade Method. If you want to save the current configuration parameters, please check **"Save current configuration"**, click **"Choose file"** to select the firmware to be upgraded through the browser version, click the **"Upgrade"** button and then click **"OK"** to confirm the upgrade. At this time, if the lights on the top of the gateway are off, it means that the gateway is being upgraded. Do not disconnect the power supply during the upgrade, wait for about 3 minutes, until the dynamic lights on the top of the gateway light up again, it means that the gateway firmware has been upgraded successfully, and the web interface is shown in the figure below.



FIRMWARE UPGRADE

Firmware Type
WiFi

Current Version: v3.2.1

Upgrade Method
FILE Upgrade

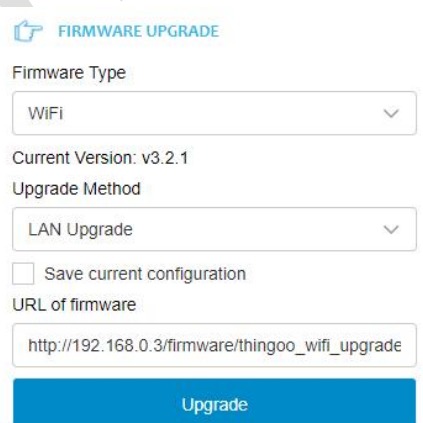
Choose file

☐ Save current configuration

Upgrade

4.4.1.3 LAN upgrade

In the case that the gateway cannot be upgraded via USB and cannot be connected to the external network, select WiFi for Firmware Type and LAN Upgrade for Upgrade Type in the Other setting bar of the web interface. If you want to save the current configuration parameters, please tick **"Save current configuration"**. Select it, fill in the URL of the firmware (as to how to build HFS, please refer to the upload certificate section of MQTT access above), click the **"Upgrade"** button and then click **"OK"** to confirm the upgrade. After confirming the upgrade, the firmware will be downloaded for a period of time. If the lights on the top of the gateway are off, it means that the gateway has finished downloading the firmware and is upgrading. Do not disconnect the power supply during the upgrade, wait for about 3 minutes, until the dynamic lights on the top of the gateway light up again, it means that the gateway firmware has been upgraded successfully, and the web interface is shown in the figure below.



FIRMWARE UPGRADE

Firmware Type
WiFi

Current Version: v3.2.1

Upgrade Method
LAN Upgrade

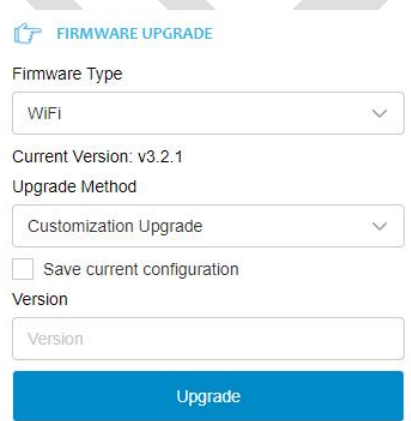
☐ Save current configuration

URL of firmware
http://192.168.0.3/firmware/thingoo_wifi_upgrade

Upgrade

4.4.1.4 Upgrade of Customized Firmware

When the gateway can be connected to the external network, in the Other setting column of the web interface, select WiFi for Firmware Type, and Customization Upgrade for Upgrade Type, and fill in the version number of the firmware customized by our company for customers. If you want to save the current configuration parameters, please Check **"Save current configuration"**, click the **"Upgrade"** button and then click **"OK"** to confirm the upgrade. After confirming the upgrade, the firmware will be downloaded for a period of time. If the lights on the top of the gateway are off, it means that the gateway has finished downloading the firmware and is upgrading. Do not disconnect the power supply during the upgrade, wait for about 3 minutes, until the top light of the gateway lights up again, it means that the gateway firmware has been upgraded successfully, and the web interface is shown in the figure below.



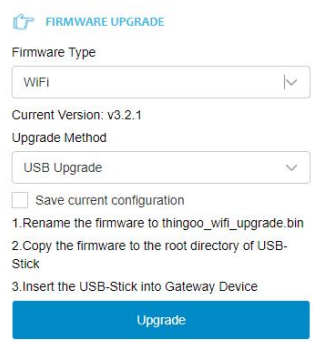
The screenshot shows a web interface titled "FIRMWARE UPGRADE". It contains the following fields and controls:

- Firmware Type:** A dropdown menu with "WiFi" selected.
- Current Version:** Displays "v3.2.1".
- Upgrade Method:** A dropdown menu with "Customization Upgrade" selected.
- Save current configuration:** An unchecked checkbox.
- Version:** A text input field with the placeholder "Version".
- Upgrade:** A blue button at the bottom.

4.4.1.5 USB Upgrade

First, rename the WiFi firmware to **"thingoo_WiFi_upgrade.bin"**, then copy the thingoo_WiFi_upgrade.bin file to the root directory of the U disk, insert the USB interface into the gateway device, and select WiFi, Upgrade Type for Firmware Type in the Other setting column of the web interface Select USB Upgrade, if you want to save the current configuration parameters, please check **"Save the configuration"**, click the **"Upgrade"** button and then click **"OK"** to confirm the upgrade. At this time, if the lights on the top of the gateway are off, it

means that the gateway is being upgraded. Do not disconnect the power supply during the upgrade, wait for about 3 minutes, until the dynamic lights on the top of the gateway light up again, it means that the gateway firmware has been upgraded successfully, and the web interface is shown in the figure below.



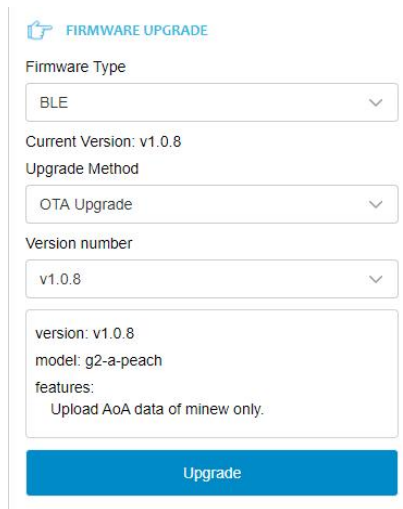
4.4.2 Bluetooth Firmware Upgrade

Bluetooth firmware upgrade supports USB upgrade, air upgrade, FILE upgrade, LAN upgrade and custom firmware upgrade (essentially air upgrade).

Note: During the upgrade process, the gateway will upgrade the firmware of all Bluetooth modules, and the upgrade can be completed in about 30 seconds when the network is unblocked.

4.4.2.1 Over-the-air Upgrade

When the gateway can be connected to the external network, select BLE for Firmware Type in the Other setting column of the web interface, select OTA Upgrade for Upgrade Method, select the version number to be upgraded, click the **"Upgrade"** button, and then click **"OK"** to confirm the upgrade. The web interface is shown in the figure below.



FIRMWARE UPGRADE

Firmware Type
BLE

Current Version: v1.0.8

Upgrade Method
OTA Upgrade

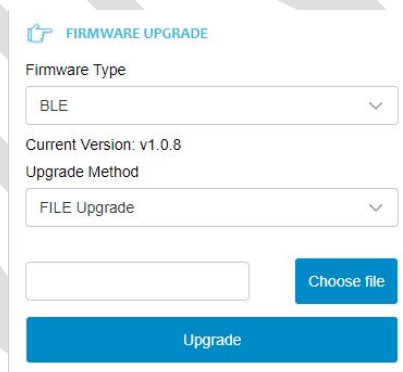
Version number
v1.0.8

version: v1.0.8
model: g2-a-peach
features:
Upload AoA data of minew only.

Upgrade

4.4.2.2 FILE Upgrade

In the Other setting column of the web interface, select BLE for Firmware Type, select FILE Upgrade for Upgrade Method, click **"Choose file"** to select the firmware version to be upgraded through the browser, click the **"Upgrade"** button, and then click **"OK"** to confirm the upgrade. The web interface is shown in the figure below.



FIRMWARE UPGRADE

Firmware Type
BLE

Current Version: v1.0.8

Upgrade Method
FILE Upgrade

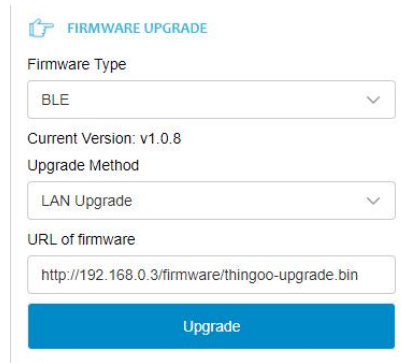
Choose file

Upgrade

4.4.2.3 LAN Upgrade

In the Other setting column of the web interface, select BLE for Firmware Type, and LAN Upgrade for Upgrade Method, and fill in the URL of the firmware (for details on how to build

HFS, please refer to the uploading certificate section of MQTT service access). Click the **"Upgrade"** button and then click the **"Upgrade"** button. Click **"OK"** to confirm the upgrade. The web interface is shown in the figure below.

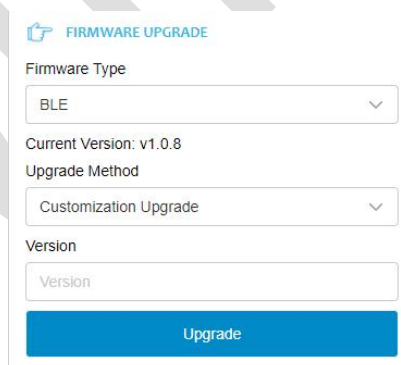


The screenshot shows a web interface titled "FIRMWARE UPGRADE". It contains the following fields and options:

- Firmware Type:** A dropdown menu with "BLE" selected.
- Current Version:** Displayed as "v1.0.8".
- Upgrade Method:** A dropdown menu with "LAN Upgrade" selected.
- URL of firmware:** A text input field containing "http://192.168.0.3/firmware/thingoo-upgrade.bin".
- Upgrade:** A blue button at the bottom.

4.4.2.4 Upgrade of Custom Firmware

When the gateway can be connected to the external network, select BLE for Firmware Type and Customization Upgrade for Upgrade Method in the Other setting column of the web interface, fill in the version number of the firmware customized by our company for customers, click the **"Upgrade"** button and then click **"OK"** to confirm the upgrade. The web interface is shown in the figure below.

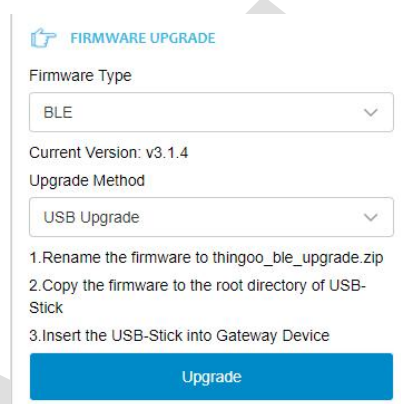


The screenshot shows a web interface titled "FIRMWARE UPGRADE". It contains the following fields and options:

- Firmware Type:** A dropdown menu with "BLE" selected.
- Current Version:** Displayed as "v1.0.8".
- Upgrade Method:** A dropdown menu with "Customization Upgrade" selected.
- Version:** A text input field with the placeholder text "Version".
- Upgrade:** A blue button at the bottom.

4.4.2.5 USB Upgrade

First rename the Bluetooth firmware to **"thingoo_ble_upgrade.zip"**, then copy the thingoo_ble_upgrade.zip file to the root directory of the U disk, insert the USB interface into the gateway device, select BLE for Firmware Type in the Other setting column of the web interface, and select BLE in the web interface. In the Other setting column of the interface, select USB Upgrade for Upgrade Type, click the **"Upgrade"** button, and then click **"OK"** to confirm the upgrade. The web interface is shown in the figure below.



The screenshot shows a web interface titled "FIRMWARE UPGRADE". It contains the following elements:

- Firmware Type:** A dropdown menu with "BLE" selected.
- Current Version:** v3.1.4
- Upgrade Method:** A dropdown menu with "USB Upgrade" selected.
- Instructions:**
 1. Rename the firmware to thingoo_ble_upgrade.zip
 2. Copy the firmware to the root directory of USB-Stick
 3. Insert the USB-Stick into Gateway Device
- Upgrade Button:** A blue button labeled "Upgrade".

4.5 Troubleshooting

To check whether the network configuration of the gateway is normal, you can use the ping and traceroute commands. The ping command is a common command to check whether the network is unblocked, and the traceroute command can display the routers that the data packets pass through to reach the destination address, so as to determine the node where the problem lies. The two are often used together.

4.5.1 Command Ping

Ping is used to determine whether the specified target address can be accessed, and to detect whether the gateway network is unobstructed.

TROUBLESHOOTING

Command
ping

Max time to execute command(seconds)
5

Number of packets sends only
1

Host/IP
www.baidu.com

Execute

OTHER

1.Reboot the gateway
Reboot

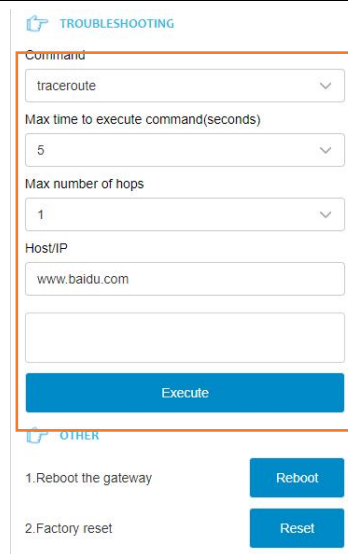
2.Factory reset
Reset

The interface fields are described in detail below.

field	describe
Command	Alternative network test commands, select the ping command
Max time to execute command	Command maximum execution time setting (5-120), in seconds, the default is 5 seconds
Number of packets sends only	The number of packets sent (1-50), the default is 1
Host/IP	The target host or address, the default is www.baidu.com

4.5.2 Command Traceroute

traceroute can check the routing status of the data packets to the destination address, so as to determine the node where the problem is.



The interface fields are described in detail below.

field	describe
Command	Alternative network test commands, select the traceroute command
Max time to execute command	Command maximum execution time setting (5-120), in seconds, the default is 5 seconds
Max number of hops	Maximum number of hops (1-30), the default is 1
Host/IP	The target host or address, the default is www.baidu.com

4.6 Restart the Gateway

Click the Reboot button in the **"Other"** column of the webpage to achieve a one-key restart, as shown in the figure below.



4.7 Factory Reset

There are two ways to restore the factory settings: through the Reset button of the gateway or through the web page configuration

According to any of the following methods, if the lights on the top of the gateway are turned off first, and then the static six-color ring lights turn on after a few seconds, it means that the factory settings have been successfully restored and the gateway is starting.

Since the previous configuration will be lost after restoring to factory settings, please operate with caution.

4.7.1 Reset Button

In the fully activated state, that is, when the gateway lights are dynamically bright, use a sharp object to press and hold the reset button of the device through the reset hole for more than 5s and then release it, and the device will be restored to factory settings.

4.7.2 Web Page Configuration

Click the **"Factory Reset"** button in the **"Other"** page to restore the factory settings, as shown in the figure below.



4.8 Brick Rescue Mode

When the gateway is powered off during the upgrade process, or the wrong firmware is flashed, the gateway may become bricked. There are two ways to save the brick, one is through the web page (mini-web server), and the other is serial port burning.

4.8.1 Mini-web Server

At present, only the 128MB memory gateways shipped after 2019.5.15 can support the rescue of bricks through the uboot webpage. If you are not sure whether to use this method, you can judge by the following methods:

1. Power off the gateway device (or unplug the power cord).
2. Use the pins to press and hold the RESET button on the gateway device without releasing it, then power up the gateway (or plug in the power cord).
3. 6 -color light to the static white light within 6 seconds , it means that the gateway can use this method to save the brick, otherwise it is not supported.

Steps:

1. Power off the gateway device (or unplug the power cord).
2. Use the network cable to directly connect the gateway network port and the computer. At this time, the network port is the LAN port. In addition to this case, **the network port can only be used as a WAN port.**
3. Use the pin to press and hold the RESET button of the gateway device without releasing it, then power up the gateway (or plug in the power cord).
4. The gateway will change from the static 6 -color light to the static white light within 6 seconds. After entering the static white light state, it means that the gateway has started the web server, and you can release the RESET button at this time .
- 5 . The uboot firmware does not support dhcp mode, so the computer should set a static ip address in advance, the parameters are as follows:
ip: 192.168.99.2 , subnet mask: 255.255.255.0 , gateway: 192.168.99.1 .
6. Enter 192.168.99.1 on the computer browser to enter the web interface of the gateway. Upload the firmware upgrade package of the WiFi module according to the prompt on the webpage, and click upgrade to complete the rescue of the brick.
- 7 . The web interface is shown in the figure below. Note that you should update the WiFi module firmware when the gateway is bricked. Do not update **U-Boot without special instructions . Do not disconnect the power supply during the firmware update process.**

Update for Minew v1.0.0

Don't power off the device during update, if everything goes well, the device will restart.

No file chosen

U-Boot is very important,if no major bugs, please do not upgrade u-boot!!!

No file chosen

4.8.2 Serial Programming

Serial programming needs to open the gateway shell, connect the serial cable and other operations. It is more complicated and is not provided for the time being. The gateway using 64MB Flash needs to save the brick in this way. If required, please contact our sales team for operating instructions.

5. Statement

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